MECHANICS APPLIED TO AEROSPACE ENGINEERING — PART I

Topic 2: Point Particle Dynamics

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Ground Truth: slides/02.PointParticleDynamics.pdf & teoria/Notes.pdf (Chapter 4)

1. Mass, Linear Momentum, and Kinetic Energy

While kinematics describes the geometry of trajectories without regard to the physical influences causing motion, **dynamics** investigates the fundamental physical causes of motion: the forces interacting between matter.

In Newtonian dynamics, every material point P is endowed with an invariant intrinsic scalar property: its **inertial mass** m_P $[\mathrm{kg}]$, which quantitatively measures its resistance to changes in its state of motion.

CORE DYNAMICAL QUANTITIES IN AN INERTIAL REFERENCE FRAME S_0

Observed from an inertial reference frame $S_0 = \{O; \mathcal{B}_0\}$, a particle of mass m_P moving at velocity $\mathbf{v}_0^P$ possesses:

- **Linear Momentum (Quantity of Motion) $\mathbf{p}_0$ (Slide 3):**

$$\mathbf{p}_0 = m_P \, \mathbf{v}_0^P \quad [\mathrm{kg \cdot m/s}]$$

  A vector quantity strictly parallel to the instantaneous velocity vector.

- **Kinetic Energy T_0 (Slide 3):**

$$T_0 = \frac{1}{2} m_P \, \|\mathbf{v}_0^P\|^2 = \frac{1}{2} m_P \, v_0^2 \geq 0 \quad [\mathrm{J = kg \cdot m^2/s^2}]$$

  A non-negative scalar quantity proportional to the square of the particle's speed v_0.

REFERENCE FRAME DEPENDENCE

Unlike mass m_P, both $\mathbf{p}_0$ and T_0 depend directly on the chosen observer's reference frame S_0. If an observer travels alongside the particle, its relative velocity is zero, and thus its measured momentum and kinetic energy vanish.

2. Newton's Three Laws of Motion

Classical dynamics is structured upon three empirical laws formulated by Sir Isaac Newton in his *Philosophiae Naturalis Principia Mathematica* (1687):

2.1 First Law (Principle of Inertia — Slide 4)

"A particle continues in its state of rest or of uniform rectilinear motion unless acted upon by a non-zero net external force."

$$\sum \mathbf{F} = \mathbf{0} \implies \mathbf{p}_0 = \text{constant} \iff \mathbf{v}_0^P = \text{constant} \iff \mathbf{a}_0^P = \mathbf{0}$$

Mathematical formulation of the First Law in an inertial reference frame

Axiomatic Importance: The First Law establishes the very existence of **inertial reference frames**: spatial reference frames in which unforced bodies do not accelerate.

2.2 Second Law (Fundamental Law of Dynamics — Slide 5)

"The time rate of change of linear momentum of a particle measured in an inertial frame is directly proportional to, and in the direction of, the resultant force acting upon it."

$$\mathbf{F} = \left.\frac{d\mathbf{p}_0}{dt}\right|_0 = \left.\frac{d(m_P \mathbf{v}_0^P)}{dt}\right|_0$$

Newton's Second Law for variable or constant mass

For a classical point particle whose mass m_P remains strictly constant in time ($\dot{m}_P = 0$):

$$\mathbf{F} = m_P \left.\frac{d\mathbf{v}_0^P}{dt}\right|_0 = m_P \, \mathbf{a}_0^P$$

where $\mathbf{F} = \sum_i \mathbf{F}_i$ is the vector sum of all concurrent forces acting on P.

2.3 Third Law (Action-Reaction Principle — Slide 6)

"To every action there is always an equal and opposite reaction: the mutual interactions between two particles are always equal in magnitude, opposite in direction, and collinear along their line of centers."

$$\mathbf{F}_{PQ} = -\mathbf{F}_{QP}, \quad \text{collinear with } (\mathbf{r}^Q - \mathbf{r}^P)$$

Action-reaction force pair between particles P and Q

3. Systems of Particles and Momentum Conservation

Consider a system composed of N point particles $\{P_1, P_2, \dots, P_N\}$. The net force on particle i decomposes into external forces $\mathbf{F}_i^{\text{ext}}$ exerted by the environment and internal interaction forces $\mathbf{F}_{ij}^{\text{int}}$ exerted by other particles within the system (Slide 7):

$$\mathbf{F}_i = \mathbf{F}_i^{\text{ext}} + \sum_{j \neq i}^{N} \mathbf{F}_{ij}^{\text{int}}$$

Total Linear Momentum and Conservation (Slide 8)

The total linear momentum of the particle ensemble is the vector sum:

$$\mathbf{p}_{\text{tot}} = \sum_{i=1}^{N} \mathbf{p}_i = \sum_{i=1}^{N} m_i \, \mathbf{v}_0^{P_i}$$

Differentiating with respect to time in the inertial frame S_0:

$$\left. \frac{d\mathbf{p}_{\text{tot}}}{dt} \right|_0 = \sum_{i=1}^{N} \mathbf{F}_i = \sum_{i=1}^{N} \mathbf{F}_i^{\text{ext}} + \sum_{i=1}^{N} \sum_{j \neq i}^{N} \mathbf{F}_{ij}^{\text{int}}$$

By Newton's Third Law, internal action-reaction pairs cancel pairwise: $\mathbf{F}_{ij}^{\text{int}} + \mathbf{F}_{ji}^{\text{int}} = \mathbf{0}$. Therefore:

$$\left. \frac{d\mathbf{p}_{\text{tot}}}{dt} \right|_0 = \sum_{i=1}^{N} \mathbf{F}_i^{\text{ext}} = \mathbf{F}^{\text{ext}}$$

PRINCIPLE OF CONSERVATION OF TOTAL LINEAR MOMENTUM

If the resultant of all external forces acting on a system vanishes ($\mathbf{F}^{\text{ext}} = \mathbf{0}$, an **isolated system**), the total linear momentum is strictly constant in time:

$$\mathbf{p}_{\text{tot}}(t) = \text{constant vector}$$

For two interacting particles P and Q in isolation: $\mathbf{p}_0^P(t) + \mathbf{p}_0^Q(t) = \text{constant}$.

4. Equations of Motion and Trajectory Integration

Newton's second law $\mathbf{F} = m_P \, \mathbf{a}_0^P$ is a second-order vector differential equation. To integrate it analytically or numerically, we project it onto coordinate bases (Slide 9).

4.1 Scalar Equations of Motion in Cartesian Coordinates

Projecting onto the canonical Cartesian basis $\{\mathbf{i}_0, \mathbf{j}_0, \mathbf{k}_0\}$:

$$\begin{cases} m_P \, \ddot{x}(t) = F_x(t, x, y, z, \dot{x}, \dot{y}, \dot{z}) \\ m_P \, \ddot{y}(t) = F_y(t, x, y, z, \dot{x}, \dot{y}, \dot{z}) \\ m_P \, \ddot{z}(t) = F_z(t, x, y, z, \dot{x}, \dot{y}, \dot{z}) \end{cases}$$

System of three coupled second-order scalar ODEs governing particle dynamics

4.2 State-Space Form and Initial Conditions (Slide 10)

The 3 second-order ODEs can be recast as a system of **6 first-order ordinary differential equations**:

$$\begin{cases} \dot{x} = v_x, & \dot{v}_x = \frac{1}{m_P} F_x(t, x, y, z, v_x, v_y, v_z) \\ \dot{y} = v_y, & \dot{v}_y = \frac{1}{m_P} F_y(t, x, y, z, v_x, v_y, v_z) \\ \dot{z} = v_z, & \dot{v}_z = \frac{1}{m_P} F_z(t, x, y, z, v_x, v_y, v_z) \end{cases}$$

Standard state-space formulation suitable for integration and numerical solvers

The general solution contains **6 integration constants**, which are uniquely determined by the **initial conditions** at $t = 0$:

$$\mathbf{r}_0^P(0) = [x_0, y_0, z_0]^T, \quad \mathbf{v}_0^P(0) = [\dot{x}_0, \dot{y}_0, \dot{z}_0]^T$$

4.3 Official Slide Benchmark Examples (Explicit Calculus & Chain Rule)

Example 1: Trajectory Under Gravity (Slide 11)

Slide 11

Direct time-integration via the Fundamental Theorem of Calculus (Barrow's Rule)

Problem Statement: A point particle P ($m_P = 1$ kg) moves under uniform downward gravity $\mathbf{F} = -m_P g \, \mathbf{j}_0$ ($g = 10$ m/s^2). At $t = 0$, the particle is launched from the origin $\mathbf{r}_0^P(0) = \mathbf{0}$ with initial velocity $\mathbf{v}_0^P(0) = 2\,\mathbf{i}_0 + 5\,\mathbf{j}_0$ m/s. Determine the velocity field $\mathbf{v}_0^P(t)$, the position trajectory $\mathbf{r}_0^P(t)$, and the explicit path equation $y(x)$.

Step 1: Horizontal Motion Along $\mathbf{i}_0$ (x-axis)

Applying Newton's second law along the x-direction where no horizontal force acts ($F_x = 0$):

$$m_P\,\ddot{x}(t) = 0 \implies \ddot{x}(t) = \frac{d\dot{x}}{dt} = 0$$

We integrate the acceleration over time from $t' = 0$ to $t' = t$ applying the Fundamental Theorem of Calculus:

$$\int_0^t \frac{d\dot{x}}{dt'}\,dt' = \int_0^t 0\,dt' \implies \Big[\dot{x}(t')\Big]_0^t = 0 \implies \dot{x}(t) - \dot{x}(0) = 0$$

Substituting the initial horizontal velocity $\dot{x}(0) = 2\ \mathrm{m/s}$:

$$\dot{x}(t) = \dot{x}(0) = 2\ \mathrm{m/s} \quad \text{(uniform rectilinear motion)}$$

Next, because $\dot{x}(t) = \frac{dx}{dt}$, we integrate velocity over time:

$$\int_0^t \frac{dx}{dt'}\,dt' = \int_0^t 2\,dt' \implies \Big[x(t')\Big]_0^t = \Big[2t'\Big]_0^t \implies x(t) - x(0) = 2t - 0$$

With $x(0) = 0$, we obtain the parametric horizontal position:

$$x(t) = 2t \quad [\mathrm{m}]$$

Step 2: Vertical Motion Along $\mathbf{j}_0$ (y-axis)

Projecting Newton's second law onto the vertical axis where weight acts downwards ($F_y = -m_P g$):

$$m_P\,\ddot{y}(t) = -m_P g \implies \ddot{y}(t) = \frac{d\dot{y}}{dt} = -g = -10\ \mathrm{m/s^2}$$

Integrating acceleration over time from $t' = 0$ to $t' = t$:

$$\int_0^t \frac{d\dot{y}}{dt'}\,dt' = \int_0^t (-10)\,dt' \implies \Big[\dot{y}(t')\Big]_0^t = \Big[-10t'\Big]_0^t \implies \dot{y}(t) - \dot{y}(0) = -10t$$

Substituting the initial vertical velocity $\dot{y}(0) = 5\ \text{m/s}$:

$$\dot{y}(t) = 5 - 10t \quad [\text{m/s}]$$

Because $\dot{y}(t) = \frac{dy}{dt}$, we integrate once more with respect to time applying the power rule of integration ($\int t'dt' = \frac{(t')^2}{2}$):

$$\int_0^t \frac{dy}{dt'}\,dt' = \int_0^t (5 - 10t')\,dt' \implies \Big[y(t')\Big]_0^t = \left[5t' - 10\frac{(t')^2}{2}\right]_0^t$$

$$y(t) - y(0) = \left(5t - 5t^2\right) - (0 - 0)$$

With $y(0) = 0$, we find the parametric vertical position:

$$y(t) = 5t - 5t^2 \quad [\text{m}]$$

Step 3: Trajectory Cartesian Curve $y(x)$
To find the geometric path independent of time, we eliminate parameter t:

$$x(t) = 2t \implies t(x) = \frac{x}{2}$$

Substituting $t(x)$ directly into $y(t)$:

$$y(x) = 5\left(\frac{x}{2}\right) - 5\left(\frac{x}{2}\right)^2 = \frac{5}{2}x - 5\left(\frac{x^2}{4}\right) = \frac{5}{2}x - \frac{5}{4}x^2$$

This equation represents a classical concave downward parabola with vertex at $x_v = 1\ \text{m}, y_v = 1.25\ \text{m}$.

Example 2: Inverse Dynamics (Slide 12)

Slide 12

Finding the resultant force from prescribed kinematics via explicit Chain Rule

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Problem Statement: What instantaneous resultant force $\mathbf{F}(t)$ must act on a particle of mass $m_P = 2$ kg so that its velocity vector evolves strictly as $\mathbf{v}_0^P(t) = 2at^2\,\mathbf{i}_0 + e^{bt}\,\mathbf{j}_0$ (where a, b are positive dimensional constants)?

Pedagogical Rationale: In inverse dynamics, the kinematics is known and we apply Newton's second law $\mathbf{F} = m_P\,\mathbf{a}_0^P = m_P\,\frac{d\mathbf{v}_0^P}{dt}$ to deduce the required force field. Because the Cartesian unit vectors $\{\mathbf{i}_0, \mathbf{j}_0\}$ are stationary in inertial frame S_0 ($\frac{d\mathbf{i}_0}{dt} = \mathbf{0}$, $\frac{d\mathbf{j}_0}{dt} = \mathbf{0}$), we differentiate each scalar component individually.

Step 1: Differentiation of the x-component (Power Rule)

$$\frac{d}{dt}\left(2at^2\,\mathbf{i}_0\right) = 2a\left(\frac{d(t^2)}{dt}\right)\mathbf{i}_0 = 2a(2t)\,\mathbf{i}_0 = 4at\,\mathbf{i}_0$$

Step 2: Differentiation of the y-component (Chain Rule)

The vertical velocity component involves a composite exponential function $f(t) = e^{bt}$. We define the intermediate variable $u(t) = bt$. Applying the **chain rule of calculus** $\frac{df}{dt} = \frac{df}{du}\frac{du}{dt}$:

$$u(t) = bt \implies \frac{du}{dt} = \frac{d}{dt}(bt) = b$$

$$\frac{d}{dt}\left(e^{bt}\right) = \frac{d(e^u)}{du}\frac{du}{dt} = e^u \cdot b = b\,e^{bt}$$

Therefore, the vertical acceleration vector component is:

$$\frac{d}{dt}\left(e^{bt}\,\mathbf{j}_0\right) = b\,e^{bt}\,\mathbf{j}_0$$

Step 3: Synthesis of Resultant Force Vector

Combining both components, the acceleration of particle P is:

$$\mathbf{a}_0^P(t) = \frac{d\mathbf{v}_0^P}{dt} = 4at\,\mathbf{i}_0 + b\,e^{bt}\,\mathbf{j}_0$$

Multiplying by the mass of the particle $m_P = 2$ kg:

$$\mathbf{F}(t) = m_P\,\mathbf{a}_0^P(t) = 2\,\left(4at\,\mathbf{i}_0 + b\,e^{bt}\,\mathbf{j}_0\right) = 8at\,\mathbf{i}_0 + 2b\,e^{bt}\,\mathbf{j}_0 \quad [\text{N}]$$

5. Power, Work, and the Kinetic Energy Theorem

5.1 Instantaneous Power (Slide 13)

The power P exerted by a force $\mathbf{F}$ on particle P moving with velocity $\mathbf{v}_0^P$ is the scalar product:

$$P = \mathbf{F} \cdot \mathbf{v}_0^P \quad [\mathrm{W} = \mathrm{J/s}]$$

5.2 The Kinetic Energy Theorem (Work-Energy Principle)

Taking the time derivative of kinetic energy $T_0 = \frac{1}{2} m_P (\mathbf{v}_0^P \cdot \mathbf{v}_0^P)$:

$$\frac{dT_0}{dt} = m_P \, \mathbf{v}_0^P \cdot \left. \frac{d\mathbf{v}_0^P}{dt} \right|_0 = m_P \, \mathbf{a}_0^P \cdot \mathbf{v}_0^P$$

Substituting Newton's second law $\mathbf{F} = m_P \mathbf{a}_0^P$:

$$\frac{dT_0}{dt} = \mathbf{F} \cdot \mathbf{v}_0^P = P$$

The time rate of change of kinetic energy equals the instantaneous power exerted by the net force

5.3 Work Done by a Force

Integrating power over the time interval $[t_1, t_2]$:

$$W_{12} = \int_{t_1}^{t_2} P \, dt = \int_{t_1}^{t_2} (\mathbf{F} \cdot \mathbf{v}_0^P) \, dt = \int_{\mathbf{r}_1}^{\mathbf{r}_2} \mathbf{F} \cdot d\mathbf{r} = \Delta T_0 = T_{0,2} - T_{0,1}$$

FUNDAMENTAL TANGENTIAL WORK CRITERION

Expressing velocity in the intrinsic Frenet-Serret frame ($\mathbf{v}_0^P = v_0 \, \mathbf{e}_t$), power becomes:

$$P = \mathbf{F} \cdot (v_0 \, \mathbf{e}_t) = F_t \, v_0$$

A force exerts non-zero power and performs work if and only if it produces tangential acceleration ($F_t \neq 0$**).** Forces perpendicular to the trajectory (normal constraints, string tension, magnetic Lorentz forces) do **identically zero work** ($W \equiv 0$).

6. Conservative Forces and Potential Energy

In general, the work integral $W_{12} = \int_\Gamma \mathbf{F} \cdot d\mathbf{r}$ depends on the path Γ connecting the endpoints.

6.1 Conservative Forces (Slide 14)

A force $\mathbf{F}$ is **conservative** if the work it performs between two points is **strictly path-independent**:

$$\int_{\Gamma_A} \mathbf{F} \cdot d\mathbf{r} = \int_{\Gamma_B} \mathbf{F} \cdot d\mathbf{r} \iff \oint_{\mathcal{C}} \mathbf{F} \cdot d\mathbf{r} = 0 \quad \text{along any closed curve } \mathcal{C}$$

6.2 Potential Energy Function $V_0(\mathbf{r})$

Fixing an arbitrary reference position $\mathbf{r}_*$, the potential energy is defined by:

$$V_0(\mathbf{r}) = -\int_{\mathbf{r}_*}^{\mathbf{r}} \mathbf{F} \cdot d\mathbf{r}'$$

The work done by a conservative force equals the negative change in potential energy:

$$W_{12} = -(V_0(\mathbf{r}_2) - V_0(\mathbf{r}_1)) = -\Delta V_0$$

Differentiating: $dW = \mathbf{F} \cdot d\mathbf{r} = -dV_0 = -\nabla V_0 \cdot d\mathbf{r}$, establishing:

$$\mathbf{F} = -\nabla V_0 = -\begin{bmatrix} \frac{\partial V_0}{\partial x} \\ \frac{\partial V_0}{\partial y} \\ \frac{\partial V_0}{\partial z} \end{bmatrix}$$

A conservative force is always the negative gradient of its scalar potential energy function

6.3 Mathematical Condition: The Curl Test (Slide 16)

In a simply-connected domain, a continuously differentiable force field $\mathbf{F}(\mathbf{r})$ is conservative **if and only if its curl vanishes identically**:

$$\nabla \times \mathbf{F} = \mathbf{0} \iff \operatorname{curl}(\mathbf{F}) = \mathbf{0}$$

6.4 Fundamental Aerospace Potential Functions (Slide 16)

Physical Force	Vector Expression F	Potential Energy $V_0(\mathbf{r})$
Uniform Gravity	$\mathbf{W} = -m_P g\,\mathbf{k}_0$	$V_0(z) = m_P g z + C$
Linear Spring (zero natural length)	$\mathbf{F} = -k\,\mathbf{r}_0^P$	$V_0(\mathbf{r}) = \frac{1}{2}k\,\|\mathbf{r}_0^P\|^2 = \frac{1}{2}k\,r^2$
Inverse-Square Central Gravity	$\mathbf{F} = -\frac{GMm_P}{r^2}\mathbf{e}_r$	$V_0(r) = -\frac{GMm_P}{r}$

7. Polar Vector Field Curl Tests

In cylindrical / polar coordinates $\{R, \theta, z\}$, the curl of a vector field is given by:

$$\nabla \times \mathbf{F} = \left(\frac{1}{R}\frac{\partial F_z}{\partial \theta} - \frac{\partial F_\theta}{\partial z}\right)\mathbf{e}_R + \left(\frac{\partial F_R}{\partial z} - \frac{\partial F_z}{\partial R}\right)\mathbf{e}_\theta + \frac{1}{R}\left(\frac{\partial (RF_\theta)}{\partial R} - \frac{\partial F_R}{\partial \theta}\right)\mathbf{e}_z$$

Example 1: Radial Force $\mathbf{F} = aR\,\mathbf{e}_R$ (Slide 17)

`Conservative`

Is it conservative? If so, compute $V_0(R)$

Evaluation: Here $F_R = aR$, $F_\theta = 0$, $F_z = 0$.

$$(\nabla \times \mathbf{F})_z = \frac{1}{R}\left(\frac{\partial (R\cdot 0)}{\partial R} - \frac{\partial (aR)}{\partial \theta}\right) = \frac{1}{R}(0 - 0) = 0$$

Since all components of $\nabla \times \mathbf{F}$ vanish identically, $\mathbf{F} = aR\,\mathbf{e}_R$ is **conservative**.

To find $V_0(R)$: $\mathbf{F} = -\nabla V_0 = -\frac{\partial V_0}{\partial R}\mathbf{e}_R \implies \frac{\partial V_0}{\partial R} = -aR$:

$$V_0(R) = -\int aR\,dR = -\frac{1}{2}aR^2 + C$$

Example 2: Azimuthal Force $\mathbf{F} = bR^2\,\mathbf{e}_\theta$ (Slide 18)

`Non-Conservative`

Is it conservative?

Evaluation: Here $F_R = 0$, $F_\theta = bR^2$, $F_z = 0$.

$$(\nabla \times \mathbf{F})_z = \frac{1}{R}\left(\frac{\partial (R \cdot bR^2)}{\partial R} - \frac{\partial (0)}{\partial \theta}\right) = \frac{1}{R}\frac{\partial (bR^3)}{\partial R} = \frac{1}{R}(3bR^2) = 3bR \neq 0$$

Because the curl is non-zero ($\nabla \times \mathbf{F} = 3bR\,\mathbf{e}_z \neq \mathbf{0}$), the force is **non-conservative** and no potential energy function can exist.

8. Mechanical Energy and 1D Energy Diagrams

Let a particle be acted upon simultaneously by conservative forces $\mathbf{F}_C = -\nabla V_0$ and non-conservative forces $\mathbf{F}_N$ (Slide 19):

$$m_P\,\mathbf{a}_0^P = \mathbf{F}_C + \mathbf{F}_N = -\nabla V_0 + \mathbf{F}_N$$

Taking the scalar product with velocity $\mathbf{v}_0^P$:

$$\frac{dT_0}{dt} = -\nabla V_0 \cdot \mathbf{v}_0^P + \mathbf{F}_N \cdot \mathbf{v}_0^P = -\frac{dV_0}{dt} + \mathbf{F}_N \cdot \mathbf{v}_0^P$$

Defining the **Total Mechanical Energy** E_0:

$$E_0 = T_0 + V_0 \implies \frac{dE_0}{dt} = \mathbf{F}_N \cdot \mathbf{v}_0^P$$

The rate of change of mechanical energy equals the power exerted by non-conservative forces

If non-conservative forces do zero work ($\mathbf{F}_N \cdot \mathbf{v}_0^P = 0$), then $E_0 = \text{constant}$ (a **first integral of motion**).

8.1 1D Qualitative Motion Analysis (Slide 20)

In a 1D conservative system:

$$T_0(x) = \frac{1}{2}m_P\,v^2 = E_0 - V_0(x) \geq 0$$

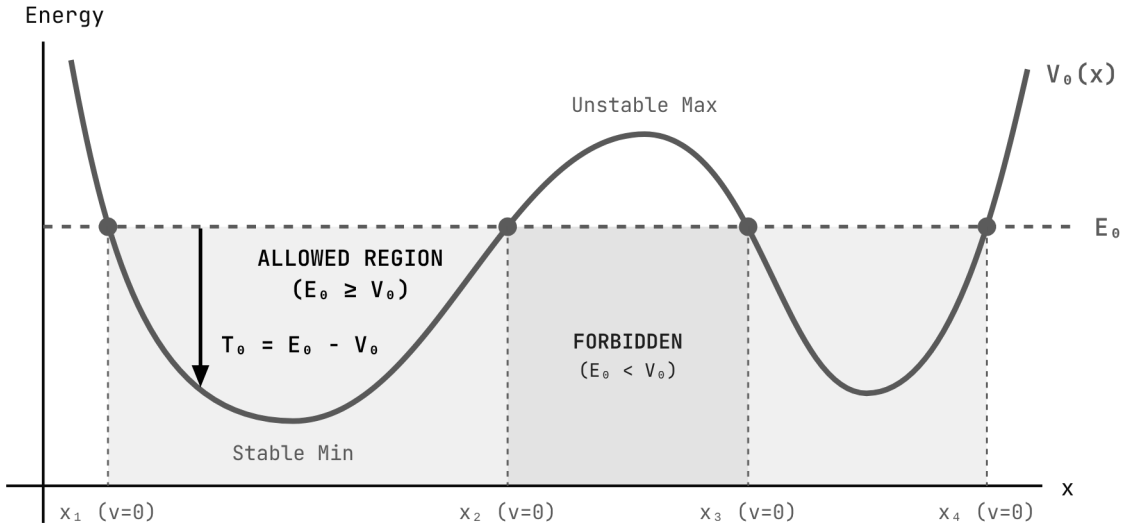


Figure 2.1: 1D energy diagram illustrating total energy E_0, potential energy $V_0(x)$, allowed regions, forbidden regions, and turning points.

- **Allowed Regions ($E_0 \geq V_0$):** Domains where physical motion can take place.
- **Forbidden Regions ($E_0 < V_0$):** Domains where kinetic energy would be negative ($v^2 < 0$), physically inaccessible.
- **Turning Points (x_{turn}):** Points where $E_0 = V_0(x_{\text{turn}}) \implies T_0 = 0 \implies v = 0$. The particle stops and reverses its direction of motion.
- **Instantaneous Speed:** $v(x) = \sqrt{\frac{2(E_0 - V_0(x))}{m_P}}$. Speed is maximal at the local minimum of $V_0(x)$.

8.2 Slide 15 Potential Example (Explicit Chain Rule Derivation)

Force Field from 1D Potential Function (Slide 15)

Slide 15

Analytical differentiation using the chain rule

Problem: Determine the 1D force $F(x)$ associated with the potential energy function:

$$V_0(x) = \cos(x) + \cos\left(\frac{x}{2} - \frac{\pi}{4}\right)$$

Pedagogical Rationale: In one dimension, a conservative force is defined by the negative spatial derivative of potential energy: $F(x) = -\frac{dV_0}{dx}$. We differentiate each term separately:

1. **First Term:** The elementary derivative of cosine is:

$$\frac{d}{dx}\big[\cos(x)\big] = -\sin(x)$$

2. **Second Term (Chain Rule):** The argument is an affine function of x. We set the intermediate variable:

$$u(x) = \frac{x}{2} - \frac{\pi}{4} \implies \frac{du}{dx} = \frac{d}{dx}\left(\frac{x}{2}\right) - \frac{d}{dx}\left(\frac{\pi}{4}\right) = \frac{1}{2} - 0 = \frac{1}{2}$$

Applying the chain rule $\frac{d}{dx}\big[f(u(x))\big] = \frac{df}{du}\frac{du}{dx}$:

$$\frac{d}{dx}\left[\cos\left(\frac{x}{2} - \frac{\pi}{4}\right)\right] = \frac{d(\cos u)}{du}\frac{du}{dx} = (-\sin u)\cdot\left(\frac{1}{2}\right) = -\frac{1}{2}\sin\left(\frac{x}{2} - \frac{\pi}{4}\right)$$

3. **Resultant Force:** Summing both derivatives and applying the fundamental leading minus sign ($F = -dV_0/dx$):

$$F(x) = -\left[-\sin(x) - \frac{1}{2}\sin\left(\frac{x}{2} - \frac{\pi}{4}\right)\right] = \sin(x) + \frac{1}{2}\sin\left(\frac{x}{2} - \frac{\pi}{4}\right)$$

Physical Interpretation: The force always acts in the direction of negative potential gradient (directing the particle "downhill" into potential valleys), with magnitude equal to the slope of the curve.

9. Equilibrium Configurations and Formal Stability Criteria

9.1 Equilibrium Configuration (Slide 21)

A configuration $\mathbf{r}_*$ is an **equilibrium configuration** if a particle released at $\mathbf{r}_*$ with zero velocity remains permanently at rest:

$$\mathbf{a}_0^P = \mathbf{0} \iff \sum \mathbf{F}(\mathbf{r}_*) = \mathbf{0} \iff \nabla V_0(\mathbf{r}_*) = \mathbf{0}$$

In 1D, equilibrium positions correspond to the critical points of the potential energy function: $\frac{dV_0}{dx}\big|_{x_*} = 0$.

9.2 Formal Definition of Stability (Lyapunov Criterion)

An equilibrium position $\mathbf{r}_*$ is **stable** if for any arbitrarily chosen position and velocity tolerances $\varepsilon_r, \varepsilon_v > 0$, there exist bounds $\delta_r, \delta_v > 0$ such that if the particle is perturbed initially with $\|\mathbf{r}(0) - \mathbf{r}_*\| < \delta_r$ and $\|\mathbf{v}(0)\| < \delta_v$, then:

$$\|\mathbf{r}(t) - \mathbf{r}_*\| < \varepsilon_r \quad \text{and} \quad \|\mathbf{v}(t)\| < \varepsilon_v \quad \forall t \geq 0$$

Otherwise, the equilibrium configuration is **unstable**.

9.3 Second-Derivative Curvature Criterion in 1D (Slide 22)

Expanding $V_0(x)$ in a Taylor series about equilibrium x_*:

$$F(x) = -\frac{dV_0}{dx} \approx -\left.\frac{d^2V_0}{dx^2}\right|_{x_*} (x - x_*)$$

- **Stable Equilibrium ($\left.\frac{d^2V_0}{dx^2}\right|_{x_*} > 0$):** x_* is a **local minimum** (potential well). The force is restoring ($F \approx -k(x - x_*)$), pushing the particle back towards x_*.
- **Unstable Equilibrium ($\left.\frac{d^2V_0}{dx^2}\right|_{x_*} < 0$):** x_* is a **local maximum** (potential crest). The force is repulsive ($F \approx +k(x - x_*)$), driving the particle away from x_*.
- **Neutral / Indeterminate Equilibrium ($\left.\frac{d^2V_0}{dx^2}\right|_{x_*} = 0$):** Inflection point; higher-order derivatives must be inspected.

10. Solved Benchmark Problem: The Simple Pendulum (Slide 23 & Notes Sec. 4.7)

ANALYTICAL & PEDAGOGICAL RIGOR AUDIT (PROBLEM_STEP_MENTOR)

This classic problem synthesizes all core concepts of Topic 2: polar kinematic constraints, dynamic force balance, string tension determination, exact mechanical energy first integral, equilibrium stability, and small-amplitude harmonic linearization.

The Simple Pendulum (Slide 23 & Notes Sec. 4.7)

UC3M Benchmark

Nonlinear dynamics, energy first integral, and harmonic linearization

Problem Statement: A point particle P of mass m is suspended from a fixed point O by an inextensible, massless rigid rod of length ℓ, subject to uniform downward gravity $\mathbf{g} = -g\mathbf{j}_0$. Determine the equations of motion, string tension, conservation of energy, equilibrium stability, and small-amplitude oscillations.

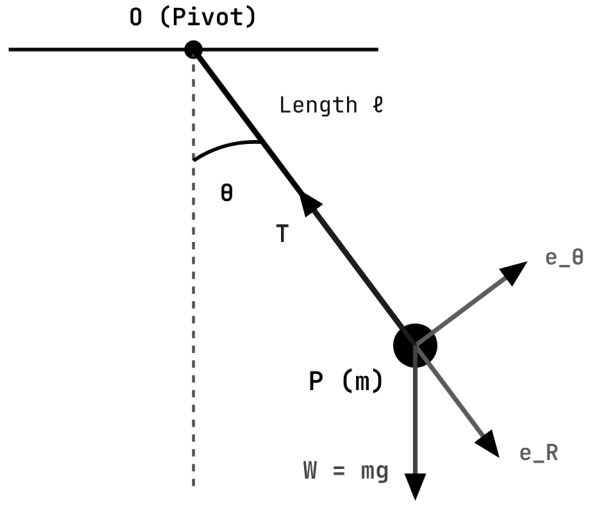


Figure 2.2: Free body diagram of the simple pendulum with polar unit vectors $\{\mathbf{e}_R, \mathbf{e}_\theta\}$, gravity $\mathbf{W}$, and tension $\mathbf{T}$.

Step 1: Coordinates, Basis, and Kinematic Constraints (Explicit Calculus)

Pedagogical Justification: The rod maintains a constant distance $R = \ell$ from pivot O, constraining the particle to a circular trajectory in the vertical Oxy plane. We define polar coordinates $\{R, \theta\}$ with fixed origin at pivot O, equipped with the local orthonormal basis $\{\mathbf{e}_R, \mathbf{e}_\theta\}$. The angle θ is measured from the downward vertical (so $\theta = 0$ is the hanging equilibrium and $\mathbf{e}_R(0) = -\mathbf{j}_0$), counter-clockwise in the Oxy plane:

$$\mathbf{e}_R(\theta) = \sin\theta\,\mathbf{i}_0 - \cos\theta\,\mathbf{j}_0$$

$$\mathbf{e}_\theta(\theta) = \cos\theta\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0$$

We differentiate the rotating basis vectors with respect to time applying the **chain rule** ($\frac{d}{dt} = \frac{d\theta}{dt}\frac{\partial}{\partial\theta} = \dot\theta\frac{\partial}{\partial\theta}$):

$$\frac{d\mathbf{e}_R}{dt} = \frac{\partial\mathbf{e}_R}{\partial\theta}\frac{d\theta}{dt} = \left(\frac{d(\sin\theta)}{d\theta}\mathbf{i}_0 - \frac{d(\cos\theta)}{d\theta}\mathbf{j}_0\right)\dot\theta = (\cos\theta\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0)\dot\theta = \dot\theta\,\mathbf{e}_\theta$$

$$\frac{d\mathbf{e}_\theta}{dt} = \frac{\partial\mathbf{e}_\theta}{\partial\theta}\frac{d\theta}{dt} = \left(\frac{d(\cos\theta)}{d\theta}\mathbf{i}_0 + \frac{d(\sin\theta)}{d\theta}\mathbf{j}_0\right)\dot\theta = (-\sin\theta\,\mathbf{i}_0 + \cos\theta\,\mathbf{j}_0)\dot\theta = -\dot\theta\,\mathbf{e}_R$$

The rigidity and inextensibility of the rod impose the holonomic constraint:

$$R(t) = \ell = \text{constant} \implies \dot{R}(t) = 0, \quad \ddot{R}(t) = 0$$

- **Position Vector:**

$$\mathbf{r}_0^P = \ell\, \mathbf{e}_R$$

- **Velocity Vector (Product Rule):**

$$\mathbf{v}_0^P = \frac{d}{dt}\left(\ell\, \mathbf{e}_R\right) = \underbrace{\frac{d\ell}{dt}}_{=0} \mathbf{e}_R + \ell \frac{d\mathbf{e}_R}{dt} = \ell(\dot{\theta}\, \mathbf{e}_\theta) = \ell\dot{\theta}\, \mathbf{e}_\theta$$

- **Acceleration Vector (Product Rule & Chain Rule):**

$$\mathbf{a}_0^P = \frac{d}{dt}\left(\ell\dot{\theta}\, \mathbf{e}_\theta\right) = \ell\left(\frac{d\dot{\theta}}{dt}\mathbf{e}_\theta + \dot{\theta}\frac{d\mathbf{e}_\theta}{dt}\right) = \ell\left(\ddot{\theta}\, \mathbf{e}_\theta + \dot{\theta}(-\dot{\theta}\, \mathbf{e}_R)\right) = -\ell\dot{\theta}^2\, \mathbf{e}_R + \ell\ddot{\theta}\, \mathbf{e}_\theta$$

Step 2: Force Projection and Newton's Second Law

Two forces act on particle P:

1. **Rod Tension $\mathbf{T}$:** Acts radially inwards toward pivot O:

$$\mathbf{T} = -T\, \mathbf{e}_R \quad (T \geq 0)$$

2. **Weight $\mathbf{W}$:** Acts vertically downwards ($\mathbf{W} = -mg\mathbf{j}_0$). We project it onto the polar basis $\{\mathbf{e}_R, \mathbf{e}_\theta\}$ via orthogonal scalar projections:

$$\mathbf{W} = (\mathbf{W} \cdot \mathbf{e}_R)\mathbf{e}_R + (\mathbf{W} \cdot \mathbf{e}_\theta)\mathbf{e}_\theta$$

$$\mathbf{W} \cdot \mathbf{e}_R = -mg\mathbf{j}_0 \cdot (\sin\theta\,\mathbf{i}_0 - \cos\theta\,\mathbf{j}_0) = mg\cos\theta$$

$$\mathbf{W} \cdot \mathbf{e}_\theta = -mg\mathbf{j}_0 \cdot (\cos\theta\,\mathbf{i}_0 + \sin\theta\,\mathbf{j}_0) = -mg\sin\theta$$

$$\mathbf{W} = mg\cos\theta\,\mathbf{e}_R - mg\sin\theta\,\mathbf{e}_\theta$$

Applying Newton's second law: $m\mathbf{a}_0^P = \mathbf{W} + \mathbf{T}$:

$$m(-\ell\dot{\theta}^2\,\mathbf{e}_R + \ell\ddot{\theta}\,\mathbf{e}_\theta) = (mg\cos\theta - T)\,\mathbf{e}_R - mg\sin\theta\,\mathbf{e}_\theta$$

Equating components along each basis vector:

- **Radial Component ($\mathbf{e}_R$):**

$$-m\ell\dot{\theta}^2 = mg\cos\theta - T \implies T(\theta,\dot{\theta}) = m(g\cos\theta + \ell\dot{\theta}^2)$$

  Physical insight: The rod tension must simultaneously offset the radial gravitational component $mg\cos\theta$ and supply the centripetal acceleration $m\ell\dot{\theta}^2$.

- **Azimuthal Component ($\mathbf{e}_\theta$):**

$$m\ell\ddot{\theta} = -mg\sin\theta \implies \ddot{\theta} + \frac{g}{\ell}\sin\theta = 0$$

  Physical insight: This is the exact, nonlinear second-order differential equation of motion of the simple pendulum.

Step 3: Mechanical Energy Conservation, Explicit Time Derivative, & First Integral

Because tension $\mathbf{T}$ acts perpendicular to the instantaneous velocity ($\mathbf{T} \cdot \mathbf{v}_0^P = (-T\mathbf{e}_R) \cdot (\ell\dot{\theta}\mathbf{e}_\theta) = 0$), non-conservative work vanishes identically. Mechanical energy E_0 is strictly conserved:

- **Kinetic Energy:** $T_0 = \frac{1}{2}m\|\mathbf{v}_0^P\|^2 = \frac{1}{2}m(\ell\dot{\theta})^2 = \frac{1}{2}m\ell^2\dot{\theta}^2$
- **Potential Energy:** Setting $V_0 = 0$ at the lowest point ($\theta = 0$): $V_0(\theta) = mg\ell(1 - \cos\theta)$
- **Total Mechanical Energy:**

$$E_0 = T_0 + V_0 = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell(1 - \cos\theta) = \text{constant}$$

Rigorous Proof of Conservation via the Chain Rule: Differentiating E_0 with respect to time t:

$$\frac{dT_0}{dt} = \frac{d}{dt}\left(\frac{1}{2}m\ell^2\dot{\theta}^2\right) = \frac{1}{2}m\ell^2\left(\frac{d(\dot{\theta}^2)}{d\dot{\theta}}\frac{d\dot{\theta}}{dt}\right) = \frac{1}{2}m\ell^2(2\dot{\theta}\ddot{\theta}) = m\ell^2\dot{\theta}\ddot{\theta}$$

$$\frac{dV_0}{dt} = \frac{d}{dt}\left[mg\ell(1 - \cos\theta)\right] = mg\ell\left(\frac{d(1-\cos\theta)}{d\theta}\frac{d\theta}{dt}\right) = mg\ell(\sin\theta)\dot{\theta}$$

Summing both terms:

$$\frac{dE_0}{dt} = m\ell^2\dot{\theta}\ddot{\theta} + mg\ell\sin\theta\dot{\theta} = m\ell\dot{\theta}\left(\ell\ddot{\theta} + g\sin\theta\right)$$

Since $\ell\ddot{\theta} + g\sin\theta = 0$ by the equation of motion, $\frac{dE_0}{dt} \equiv 0$ identically!

Solving $E_0 = \frac{1}{2}m\ell^2\dot{\theta}^2 + mg\ell(1 - \cos\theta)$ for angular velocity $\dot{\theta}$ yields the **first integral of motion**:

$$\dot{\theta} = \pm\sqrt{\frac{2}{m\ell^2}\left[E_0 - mg\ell(1 - \cos\theta)\right]}$$

Turning points occur when $\dot{\theta} = 0 \implies \cos\theta_{\max} = 1 - \frac{E_0}{mg\ell}$.

Step 4: Equilibrium Configurations and Curvature Stability

Equilibrium positions correspond to the critical points of potential energy $V_0(\theta)$:

$$\frac{dV_0}{d\theta} = mg\ell\sin\theta = 0 \implies \theta_* = 0, \pm\pi$$

We classify stability by evaluating the second derivative (curvature test):

$$\frac{d^2V_0}{d\theta^2} = \frac{d}{d\theta}(mg\ell\sin\theta) = mg\ell\cos\theta$$

- **At $\theta_* = 0$ (hanging vertically downwards):**

$$\left.\frac{d^2V_0}{d\theta^2}\right|_{\theta=0} = mg\ell\cos(0) = +mg\ell > 0 \implies \textbf{Stable Equilibrium (Local Minimum)}$$

- **At $\theta_* = \pm\pi$ (inverted vertically upwards):**

$$\left.\frac{d^2V_0}{d\theta^2}\right|_{\theta=\pm\pi} = mg\ell\cos(\pm\pi) = -mg\ell < 0 \implies \textbf{Unstable Equilibrium (Local Maximum)}$$

Step 5: Small-Amplitude Linearization ($\theta \ll 1$)

The Maclaurin series expansion of $\sin\theta$ about the stable equilibrium $\theta = 0$ is:

$$\sin\theta = \sum_{n=0}^{\infty}\frac{(-1)^n}{(2n+1)!}\theta^{2n+1} = \theta - \frac{\theta^3}{3!} + \frac{\theta^5}{5!} - \cdots = \theta - \frac{\theta^3}{6} + \mathcal{O}(\theta^5)$$

For small angular deflections $|\theta| \ll 1$ rad (e.g., $|\theta| < 10^\circ \approx 0.175$ rad), the cubic term is negligible ($\frac{\theta^3}{6} \approx 0.00089 \ll 0.175$). Retaining only the first-order linear term $\sin\theta \approx \theta$, the nonlinear equation reduces to:

$$\ddot{\theta} + \frac{g}{\ell}\theta = 0 \iff \ddot{\theta} + \omega_0^2\theta = 0$$

Linearized second-order ODE of the simple harmonic oscillator

The characteristic polynomial is $r^2 + \omega_0^2 = 0 \implies r = \pm i\omega_0$, yielding the general harmonic oscillation:

$$\theta(t) = \theta_{\max}\cos(\omega_0 t + \phi_0)$$

The natural angular frequency ω_0 and period of oscillation T_0 are:

$$\omega_0 = \sqrt{\frac{g}{\ell}} \quad [\mathrm{rad/s}], \quad T_0 = \frac{2\pi}{\omega_0} = 2\pi\sqrt{\frac{\ell}{g}} \quad [\mathrm{s}]$$

Notice that period T_0 depends strictly on pendulum length ℓ and gravitational acceleration g, exhibiting complete independence from particle mass m and amplitude $\theta_{\max}$ (isochronism of small oscillations).